

Association Between Mechanical Power and 28-Day Mortality in Mechanically Ventilated ICU Patients: A Retrospective Cohort Analysis of the MIMIC-IV Database

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Introduction

- ❖ Invasive mechanical ventilation is lifesaving but can cause ventilator-induced lung injury (VILI).
- ❖ Despite lung-protective ventilation, ICU mortality remains high; capping single parameters leaves part of the injury unaddressed.
- ❖ Mechanical power (MP) combines tidal volume, driving pressure, respiratory rate, flow, and PEEP into one number: the energy delivered to the lung per minute (J/min).
- ❖ This study quantifies the independent association between dynamic MP delivered in the first 24 h of ventilation and 28-day all-cause mortality across a large, heterogeneous ICU cohort, and tests whether oxygenation, lung size, or age modify it.

Methodology

Design & data

- ❖ Retrospective cohort using MIMIC-IV; exposure from the PhysioNet Temporal Respiratory Support dataset.

Population (see flow chart)

- ❖ Adults on invasive ventilation (confirmed by intubation timestamps) with ≥ 4 h of computable MP in the first 24 h.
- ❖ **50,920 patients** → 18,980 analytic cohort (15,552 survivors; 3,428 non-survivors).

Exposure

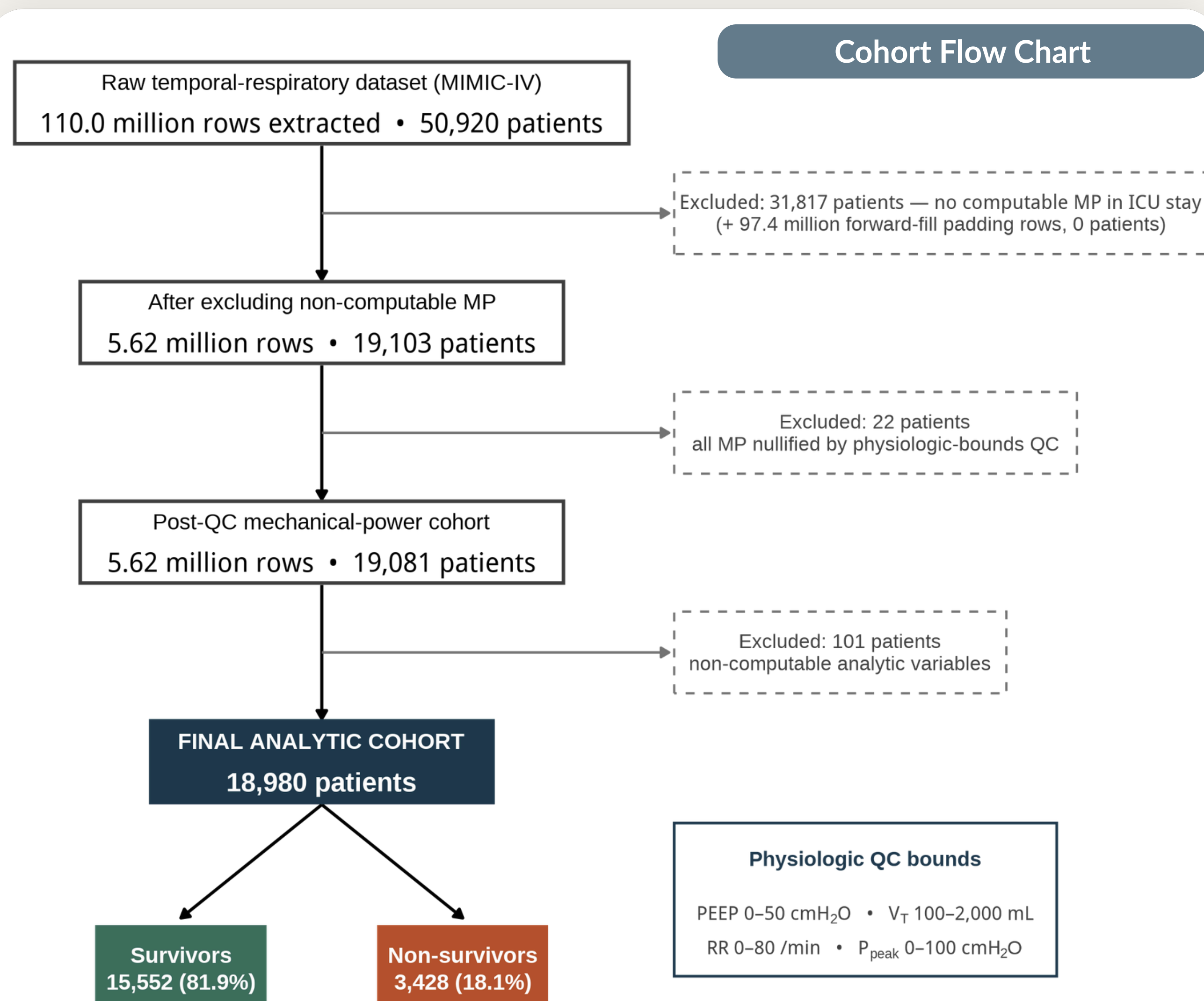
- ❖ Dynamic mechanical power (MP) over the first 24 h, as the time-weighted average (TWA).
- ❖ $MP \text{ (J/min)} = 0.098 \times RR \times V_T \times [P_{\text{peak}} - \frac{1}{2}(P_{\text{peak}} - PEEP)]$.

Outcome

- ❖ 28-day all-cause mortality (anchored to intubation).

Analysis

- ❖ Multivariable logistic regression adjusted for age, sex, predicted body weight (PBW), oxygenation (PF/SF), vasopressors, CRRT, and pulmonary phenotype.
- ❖ Restricted cubic spline for the dose-response; pre-specified subgroup and interaction analyses (oxygenation, lung size, age).



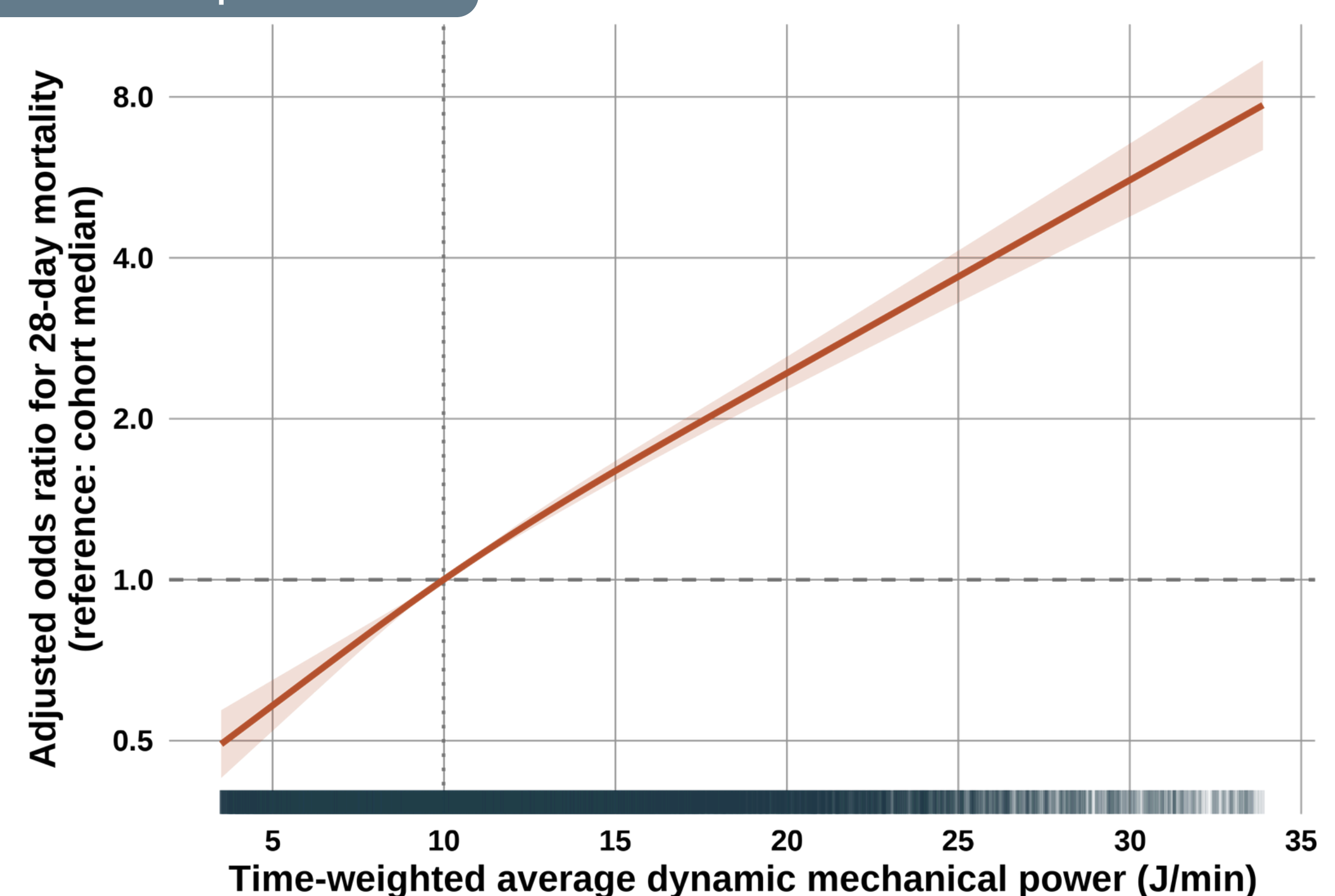
Results

Cohort comparison

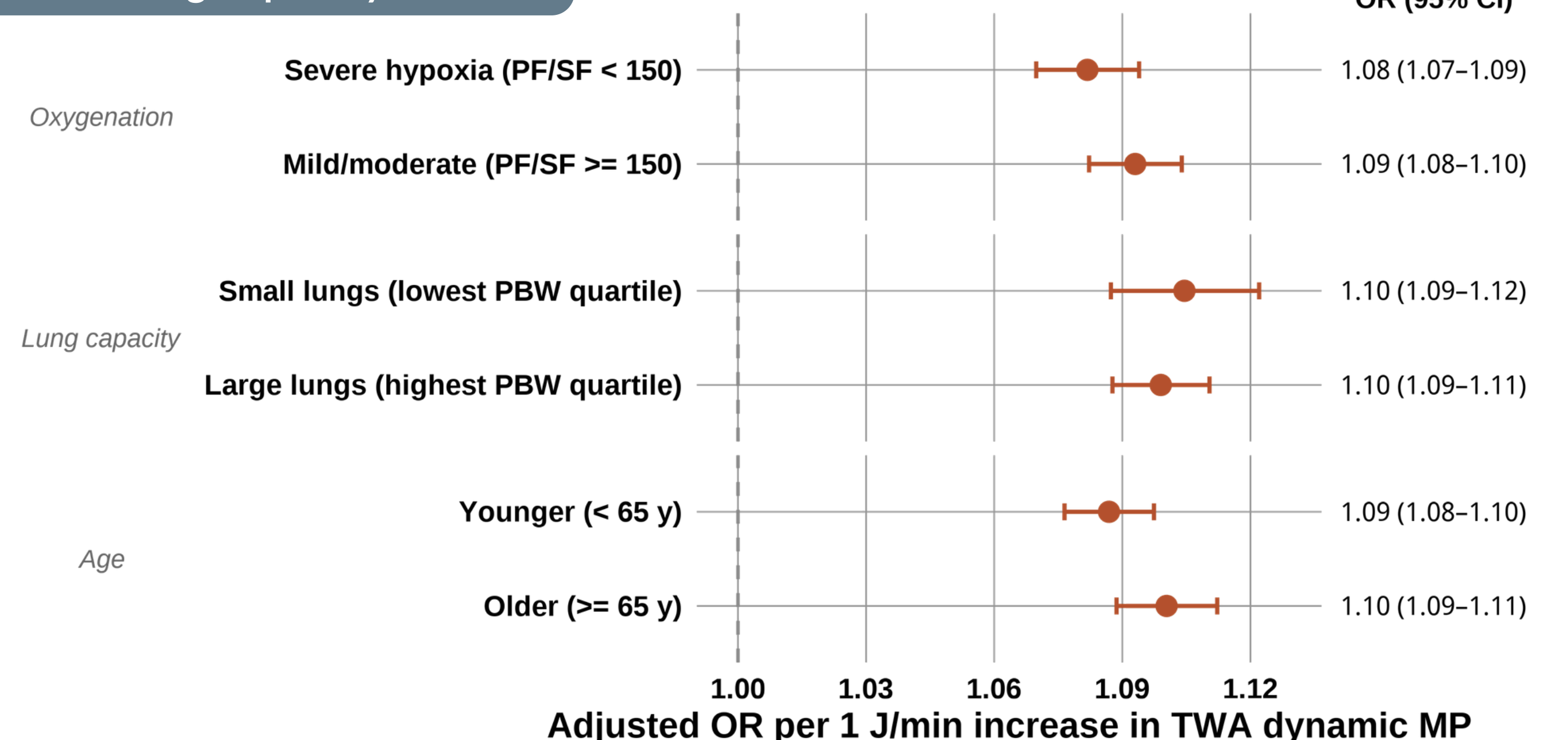
Characteristics	Survivor N = 15,552	Non-survivor N = 3,428	p-value
Age, y	64 (53-74)	71 (58-81)	< 0.001
Male sex	9,740 (63%)	1,883 (55%)	< 0.001
PBW, kg	66 (55-71)	64 (52-71)	< 0.001
Oxygenation ratio (PF/SF)	218 (175-283)	196 (125-250)	< 0.001
Vasopressors	7,917 (51%)	2,440 (71%)	< 0.001
CRRT	447 (3%)	483 (14%)	< 0.001
TWA dynamic MP, J/min	9.6 (7.4-12.8)	12.0 (8.8-18.1)	< 0.001

Median (Q1-Q3) or n (%). Wilcoxon rank-sum / chi-squared test. PF/SF = PaO₂/FIO₂ or SpO₂/FIO₂.

Dose-response



Subgroup analysis



- ❖ Of 18,980 mechanically ventilated adults, 3,428 (18.1%) died within 28 days. **Each 1 J/min increase in dynamic mechanical power was independently associated with a 9% rise in the adjusted odds of death (OR 1.09, 95% CI 1.09–1.10), and this risk was not confined to any subgroup.**
- ❖ Non-survivors received higher mechanical power than survivors (median 12.0 vs 9.6 J/min; p < .001).
- ❖ **Risk rose linearly with mechanical power**, crossing the cohort median (10 J/min) and continuing to climb with no apparent threshold (p-nonlinearity 0.067).
- ❖ **The association persisted across every pre-specified subgroup:** oxygenation, lung size, and age (OR 1.08–1.10; no significant interaction).

Conclusion

- ❖ Dynamic MP in the first 24 h is independently and consistently associated with 28-day mortality — **each 1 J/min raised the adjusted odds of death by ~9%** across a heterogeneous ICU population.
- ❖ **The MP–mortality relationship is graded and dose-dependent rather than threshold-based**, risk increases progressively with power across the studied range.
- ❖ **The association was consistent across all subgroups** (oxygenation, lung size, and age) with no significant effect modification.
- ❖ MP should be monitored at the bedside as a complement to tidal-volume, plateau-pressure, and driving-pressure targets.